ITU-T's NGN Global Standards Initiative Leading NGN Forward

NEXT GENERATION NETWORK
NGN-GSI
GLOBAL STANDARDS INITIATIVE
ITU-T

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Outline

- Introduction to ITU/ITU-T
- Networks in transition
- ITU and NGN
- ITU and cooperation
- ITU and development

Introduction to ITU

- International partnership of 191 governments and over 700 private sector entities
- Founded in 1865, it is oldest intergovernmental organisation
- ITU named in Booz Allen Hamilton survey as one of the **world's ten most enduring institutions** that have changed and grown in unswerving success and relevance — yet remained true through time to its founding principles
- Standards making one of the ITU's first activities
- Headquarters Geneva, 11 regional offices

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ITU Mission and More

- Maintain and extend international cooperation in telecommunications
- Technical and policy assistance to developing countries
- To harmonize actions of Member States and promote cooperation between Member States and Sector Members
- Leading managerial role in **World Summit on the Information Society (WSIS)**, held in Geneva Dec 2003 and Tunis Nov 2005
- Responsible for WSIS follow-up action on infrastructure and cyber security

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Historical Highlights: Standards

- 2005: VDSL2
- 2004: NGN Focus Group established
- 2004: Standards for Gigabit passive optical network (PON)
- 2002: H.264/MPEG-4/AVC next generation video coding
- 2000: X.509 for public key infrastructure (PKI) updated
- 1999: Standards for cable modems
- 1998: V.90 modem standard
- 1998: SDH – key standard for digital information
- 1996: H.323 for VoIP and videoconferencing
- 1993: First DSL standards for broadband
- 1988: Key audio coding standards (G.711 and G.72x)
- 1988: 1981: Signaling system seven (SS7) standards
- 1976: First packet switching standard (X.25)
- 1968: First standards for fax transmission

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ITU-T hot topics

- **Next generation network (NGN)**
- Bridging the standardization gap
- IPTV
- Cybersecurity including identity management (IdM)
- Ubiquitous networks
- Next generation multimedia conferencing
- Videocoding
- Broadband access
- Packet based transport
- Fibre optics
- Home networking
- ICTs and climate change

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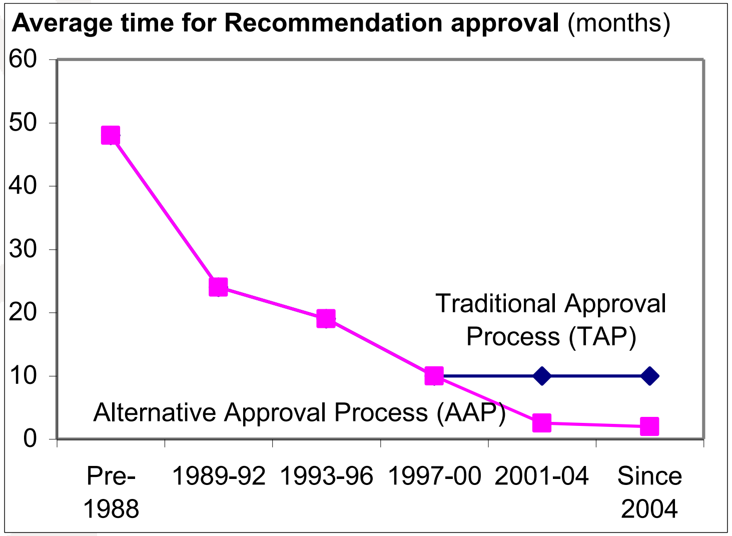
2006 Output

- 231 Recs/supplements published
- Increase of 30% over 2005 in terms of meeting documents processed
- 61 meetings (44 in Geneva, 17 outside) increase of 13% over 2005
- Strong NGN Focus
  - Global NGN standards progressing rapidly within ITU-T
  - Providing global leadership through overall framework and structure
  - Building on the work of other bodies (not reinventing ...)
  - 22 Recommendations on NGN
    - Transition from legacy, requirements, architecture, security, QoS, OAM (Ethernet and MPLS)

ITU-T Characteristics

- Quick to respond and adapt to market pressure: contribution-driven
- Unique partnership of private sector and government
- Today, 95% of standards developed and agreed by private sector
- Truly global and impartial
- Consensus decisions
- Very flexible
- Fast procedures, transparent procedures
  - start work: 1 day / few weeks
  - develop work: weeks to 2-3 yrs
  - approve work: average 2 months
- Common IPR Policy with ISO and IEC



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ITU-T Focus Groups

- Provide quick reaction to standardization needs and giving participation and working method flexibility
    - Forum-like entities with "arms-length" organization
    - A high degree of independence, adopt own working methods
    - Conclude working normally within 12 months
    - Non-ITU members can participate and can benefit from:
        - Worldwide visibility
        - Networking opportunities
        - Exposure to a large pool of expertise drawn from related work under progress in ITU-T Study Groups and other ITU-T Focus Groups.

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Upcoming workshops

- **Can we Win the War Against Cyber-Threats:** 12 November, Rio de Janeiro, Brazil
- **Making accessibility a reality in emerging technologies**: 13 November, Rio de Janeiro, Brazil
- **Towards International Standards for a Truly Multilingual Global Internet**: 13 November, Rio de Janeiro, Brazil
- **Human exposure to electromagnetic fields (EMFs)**: 20 November, Geneva, Switzerland
- **Broadband Wireless Access Seminar** 26 November, Moscow, Russia
- **The Fully Networked Car**: 5-7 March 2008, Geneva Motor Show – one of the world's top five motor shows, Switzerland

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Innovations in NGN
12-13 May 2008

- Part of initiative to increase dialogue between academia/ITU
- Follows January 2007 consultation meeting
- First in a series of '**Kaleidoscope**' events
- Will identify new topics for standardization
- IEEE Communications Society technical co-sponsor
- Open to all (paper selection via call for abstracts)
- itu.int/ITU-T/uni/kaleidoscope/

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Networks in Transition

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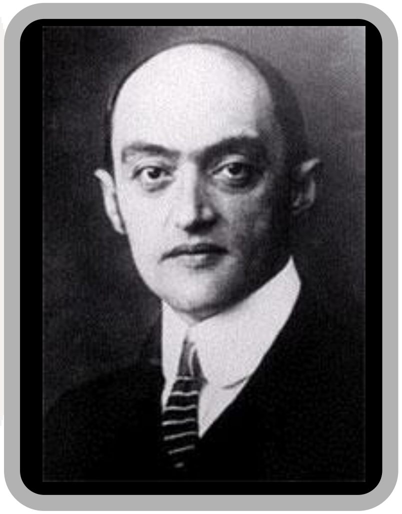
Networks in Transition: The Impact of New Communications Technologies

- Technology-driven industries like the communications sector have historically been characterized by steady growth punctuated by "giant leaps" forward, usually when "new" technology is introduced

- "Technology is not kind. It does not wait. It does not say please. It slams into existing systems. Often destroying them, while creating new ones"
  - Economist, Joseph Alois Schumpeter (1937)



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Communications revolutions

- 1840s: Telegraph
- 1870s: Telephone
- 1890s: Radio telegraphy or "wireless"
- 1920s: Radio broadcasting
- 1950s: Television broadcasting
- 1960s: Geostationary satellite communications
- 1970s: Computer communications
- 1980s: Optical communications
- 1990s: Internet and mobile
- 2000s: IP-enabled NGNs or Next Generation Internet?

1865: ITU Created

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Just How Fast Things Are Changing…

- 1998: Few ITU IP-based networks activities

  >> PP Res 101 (Minneapolis, 1998) calls upon ITU to *"fully embrace the opportunities that arise from the growth of IP-based services"*

- 2007: Almost all ITU's day-to-day activities are related to IP-based networks or the Internet
- Examples:
  - New ITU standards (DSL, cable, FTTx) have brought broadband to over 280 million new users since 2000
  - IP-enabled Next Generation Networks (NGNs)
  - IPTV, ENUM, IPv6 deployment, IDNs, cybersecurity, countering spam, IP Policy Manual, IP interconnection policies, IXPs, convergence & regulatory policies etc

Approaching tipping point?

- Deployment of new communications technologies is typically a series of relatively short cycles of one or two decades’ duration:
  - beginning with invention
  - early stages of rapid innovation and application
  - typically over-hyped and not used for original purpose intended
    - took 30 years for the “killer app” of the telephone to emerge (chat)
    - WWW took time for business models to emerge
    - SMS operators struggled to market at first
  - finally deployed in way to scale to broader market acceptance and commoditization

Next Generation Network

ITU-T Definition of NGN (Y.2001)

- Next Generation Network (NGN): a packet-based network able to provide telecommunication services and able to make use of multiple broadband, QoS-enabled transport technologies and in which service-related functions are independent from underlying transport-related technologies.
- It enables unfettered access for users to networks and to competing service providers and/or services of their choice.
- It supports generalized mobility which will allow consistent and ubiquitous provision of services to users.

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Standards: The business case

- World's ICT business asked for ITU to take lead in NGN standards in 2004
- Harmonization across boundaries increasingly important: consistent user experience
- Revenues increasingly driven by content and services rather than by type of network
- Meeting diverse and customer-segment-specific markets requires a range of solutions to interoperate
- BT aims for annualized cost savings of £1bn pa from 21st century network (21CN)
- *NTT's CEO Norio Wada speaking at ITU-T 50th anniversary, 2006: "…considering NGN performance requirements, we need de jure standards for that network. This need makes ITU-T's role even more critical. Without ITU-T's work in this area, telecoms cannot build global NGN as robust and reliable as conventional networks. I have high hopes that ITU-T will effectively address this essential need.…"*

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ITU's NGN-GSI

- GSI = Global Standards Initiative
- Developing the detailed standards necessary for NGN deployment to give service providers the means to offer the wide range of services expected in NGN.
- NGN-GSI harmonizes, in collaboration with other bodies, different approaches to NGN architecture worldwide.
- Brings together ITU-T Study Groups working on NGN standards
  - SG 13 Lead SG on **NGN**
  - SG 19 Lead SG on **mobility**
  - SG 11 Lead SG on **signalling and protocols**
  - SG 2 Service provision, networks and performance
- Co-located meetings

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Also…

- Service requirements, features, architecture, and implementation scenarios of IMS based real-time conversational multimedia services
- Requirements for Fixed Mobile Convergence
- Protocols for QoS in NGN – six Recommendations
- IdM-GSI – Identity management
- IPTV-GSI
- Advanced multimedia system (AMS)

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Collaboration

ITU and Referencing

- ITU does not do alone
  - ITU-T Recs. A.4, A.5, A.6 provide mechanism to share results
  - Process used in many Recommendations
    - E.g., IMT-2000 Family Member specs transposed by 3GPP & 3GPP2 OPs globalized in Q.1741- & Q.1742-series

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Relationship with other standards bodies

- MoUs and cooperation agreements with over 70 standards bodies

- **3GPP**: Iterative process between ITU-T and 3GPP agreed. September, 2007 meeting between TSB and 3GPP management

    - ITU-T invitation to host 3GPP meetings
    - 3GPP as an ITU-T Focus Group?
    - ITU-T as 3GPP Partner?

- **IEEE ComSoc:** MoU work in progress
- **IETF:** ITU-T / IETF Leadership Gathering, 21 July 2007
    - Area Directors and Study Group chairs

Global Standards Collaboration

- GSC 12, Kobe 8-13 July 2007
- The mandate of GSC is to provide a venue for the leaders of the Participating Standards Organizations and the ITU to:
  - ...exchange information on the progress of standards development…
  - Collaborate in planning future standards development to gain synergy and to reduce duplication.
  - …the mandate of GSC is to provide a venue… to:
  - Support the ITU as the preeminent global telecommunication and radiocommunication standards development organization.

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WSC Members have aligned their patent policies

- WSC = World Standards Cooperation (ISO, IEC and ITU)
- Common IPR Policy based on ITU-T policy
- … strongly encourages the disclosure of known patented technology from the outset.
- Allows for companies' IPR to be included in standards as long as it is made available under reasonable and non-discriminatory terms and conditions.
- WSC also adopted Guidelines for the Implementation of the Common Patent Policy and a Patent Statement and License Declaration Form.
- Each of the three WSC organizations also has an online patent database.

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ITU and Development

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Global Standards Symposium

- To be held one day before next ITU World Telecommunication Standardization Assembly (WTSA-08), in South Africa, on 20 October 2008
- Aims to bring together all standards makers
- Themes:
  - Reducing the standardization gap
  - Improving collaboration
- Details discussed at: tsagroundtable@itu.int

Bridging the standardization gap - what does it mean?

- The standardization gap might be defined as disparities in the ability of representatives of developing countries, relative to developed ones, to **access, implement, contribute to and influence international ICT standards**, specifically ITU Recommendations.

- The standardization gap is itself both **a cause and a manifestation of the wider digital divide**

- It contributes to the ***persistence*** of the wider digital divide

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What has been done recently to reduce the standardization gap?

- First in series of annual Forums in each region organised by TSB, BDT, Regional Office and Regional Organisation
- Placing greater emphasis on implementation guidelines – exploring possible translation in regions?
- Starting a series of Tutorial Groups on implementation
- Directors TSB and BDT written to all Member States and Sector Members inviting contributions for fund on bridging standardization gap
- Establishing Regional Groups
- Trialling collaborative tool for remote participation in meetings
- More meetings in regions

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Bridging the standardization gap Forums

- Addressing manufacturers, operators, service providers, regulators and administrations.
- Providing an overview of technologies that have created major standardization challenges such as NGN, multimedia, VoIP, IPTV, security and regulation.
- Highlighting ways and means to enhance cooperation and participation in ITU's standardization work and standards implementation in developing countries.
- One Forum in each region – Asia Pacific, Americas, Arab, Africa - before October 2008

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Free Recommendations

- Historic decision taken by the 2007 Council
- Major step forward in bridging the standardization gap
- During trial Jan-August 2007 16% of downloads to developing countries
- 300’000 downloads to developing countries during 8 month trail, compared with just 500 sold in 2006
- Total of 2 million downloads during trial

NGN-GSI Key Recommendations

Rec. N.	Title
Y.2012	Functional requirements and architecture of the NGN
Y.2091	Terms and definitions for Next Generation Networks
Y.2111	Resource and admission control functions in NGN
Y.2171	Admission control priority levels in Next Generation Networks
Y.2021	IMS for Next Generation Networks
Y.2261	PSTN/ISDN evolution to NGN
Y.2031	PSTN/ISDN emulation architecture
Y.2271	Call server based PSTN/ISDN emulation
Y.2201	NGN release 1 requirements
Y.2701	Security requirements for NGN release 1
Q.1706/Y.2801	Mobility management requirements for NGN
Q.3900	Method of testing and model network architecture for NGN Technical means testing as applied to public telecommunication networks
M.3343	Requirements and analysis for NGN trouble administration across B2B and C2B interface
Y.1542	Framework for achieving end-to-end IP performance objectives

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Conclusion

- **ITU the only international, intergovernmental organization developing telecommunication/ICT standards.**
- Global NGN standards progressing rapidly within ITU-T
  - Providing global leadership through overall framework and structure
  - Taking advantage of the work of other bodies (not reinventing ...)
  - Extending benefits to developing world
- Collaboration to avoid duplication of effort and incompatible implementations is essential

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